

NETFLIX

ORIGINAL PROJECT

Reimagining Content Discovery

Building a Scalable Personalized Recommendation Engine

- Netflix Prize Dataset (100M+ ratings)
- Large-scale Collaborative Filtering
- Explainable Recommendation Pipeline

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100M+

Movie Ratings

SVD

Prize-Winning
Algorithm

Netflix Prize Dataset

Collaborative Filtering · Matrix Factorization · ALS · SVD

RMSE

0.98

Best Score

TOP 10

**Personalized
Recs**

Problem Statement

Why Recommendation Systems Are Challenging

KEY OBJECTIVE

Predict user preferences and recommend Top-10 movies efficiently.

01

100M+ Ratings

Over 100 million user-movie ratings across thousands of titles to process.

02

Extreme Sparsity

User-movie matrix is extremely sparse — most users rate <1% of movies.

03

Prediction Accuracy

Need accurate predictions at scale with minimal error (RMSE).

04

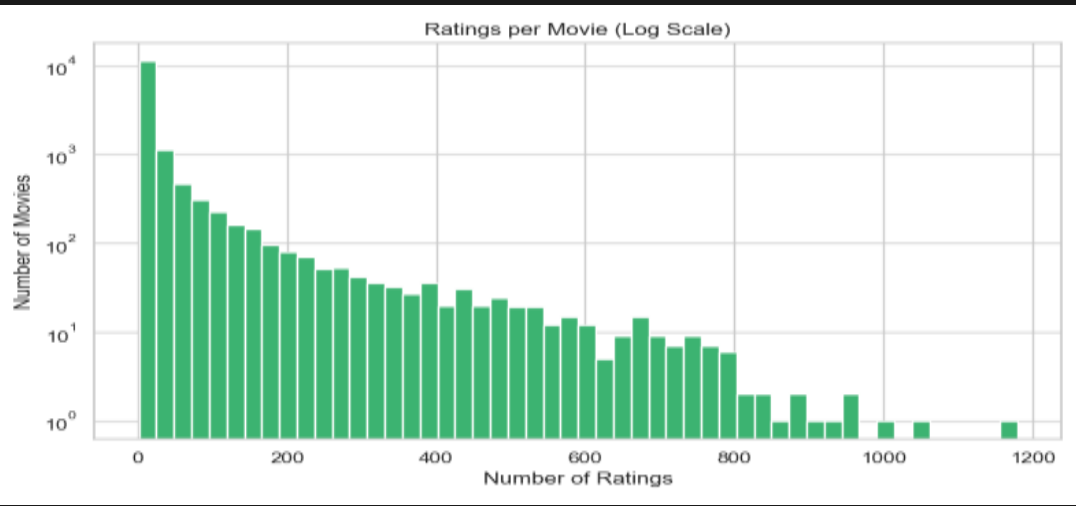
Memory Constraints

Must process large datasets on limited computational resources.

VISUAL: Sparse matrix illustration — user × movie grid with mostly empty cells

Dataset Insights

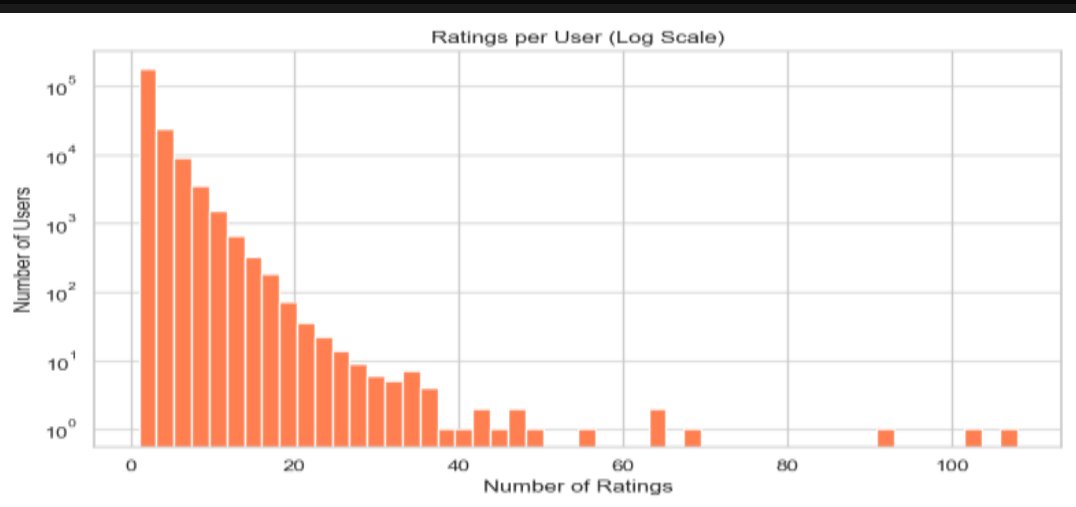
Understanding User Behavior



POSITIVITY BIAS

Users prefer 3 to 5 stars

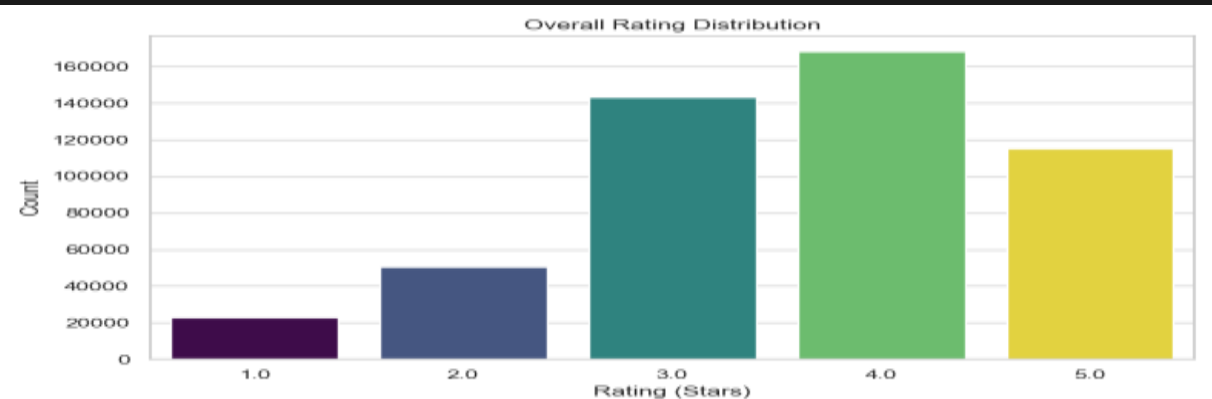
Ratings of 4 ★ dominate — strong optimistic skew in the dataset.



THE LONG TAIL

Few movies dominate interactions

Thousands of niche movies receive very little interaction — the long-tail effect.



TAKEAWAY: Recs must balance popularity and personalization

Data Engineering Pipeline

Processing Large-Scale Data Efficiently

RAW DATA

100M+ ratings
2 GB text files

PARQUET FORMAT

10x compression
Fast I/O reads

TEMPORAL SPLIT

80/20 train/test
Date-ordered

SPARSE MATRIX

SciPy CSR format
Memory efficient

ML READY

CF · ALS · SVD
Pipeline done

TEMPORAL VALIDATION — No Data Leakage

- Ratings sorted chronologically by date
- 80% training / 20% test split
- Future data never leaks into training
- Mirrors real-world deployment constraints

MEMORY OPTIMIZATION — Scale on Any Hardware

- Converted raw text to Parquet format
- SciPy sparse matrices (CSR) used
- Memory footprint reduced significantly
- Runs on standard consumer hardware

Recommendation Models

Algorithms Evaluated — Three Approaches, One Winner

Item-Based CF

Collaborative Filtering

Recommends movies that are mathematically similar to ones the user already liked.

- ✓ Cosine similarity between item vectors
- ✓ Intuitive & interpretable results
- ✓ No training needed — instant lookup
- ⚠ Struggles with very sparse data

ALS

Alternating Least Squares

Purpose-built for sparse datasets. Iteratively solves user/item latent factors.

- ✓ Designed for sparse matrix data
- ✓ Efficient matrix factorization
- ✓ Scalable and parallelizable
- ⚠ Requires iteration tuning

SVD

Singular Value Decomposition

Learns hidden user and movie latent features — capturing deep preference patterns.

- ✓ Discovers latent preference factors
- ✓ Captures hidden content patterns
- ✓ Best overall prediction accuracy
- ⚠ RMSE = 0.98 on test set

🏆 NETFLIX PRIZE WINNER

Performance Evaluation

Benchmarking the Models — RMSE & MAP@10

RMSE — Root Mean Square Error

Measures rating prediction accuracy.
Lower score = better prediction.

MAP@10 — Mean Average Precision @ 10

Measures relevance of top-10 recommendations
against what user actually watched.

MODEL COMPARISON

MODEL	RMSE ↓	MAP@10 ↑	PERFORMANCE VERDICT
Item-Based CF	1.24	0.031	Baseline — interpretable but limited
ALS	1.08	0.047	Good — handles sparsity well
SVD ★ Winner	0.98	0.061	Best — Prize-winning algorithm

🏆 SVD — RMSE 0.98 | MAP@10 0.061 — Best Overall Performance

Successfully learned hidden user and movie latent features. The same algorithm that won the original Netflix Prize competition.

Interactive Recommendation Dashboard

Explainable AI for Recommendations — Built with Streamlit

• • •

localhost:8501 — NetflixRecommender | Streamlit

🎬 Recommender

› User ID

› Algorithm

› Top-K

› Explain

🏆 Your Top 10 Recommendations

1

The Shawshank Redemption

★★★★★

2

Schindler's List

★★★★★

3

The Dark Knight

★★★★☆

4

Pulp Fiction

★★★★☆

5

Forrest Gump

★★★★☆

🔍 Why was this recommended?

Extracts SVD latent factors to mathematically explain similarity. No black box.

📊 Latent Factor Explainability

Visualizes the hidden genre/preference dimensions that drove each recommendation.

💬 Example Explanation

"Because you liked Movie A & B, Movie C shares similar latent features."

Conclusions & Future Work

PROJECT OUTCOMES

- ✓ Processed 100M+ ratings efficiently on standard hardware
- ✓ Built a scalable end-to-end recommendation pipeline
- ✓ SVD achieved best-in-class RMSE of 0.98
- ✓ Generated explainable recommendations via Streamlit
- ✓ Full latent-factor transparency — no black box

FINAL TAKEAWAY

Scalable ML can transform content discovery through personalized recommendations.

FUTURE DIRECTIONS

The Road Ahead

🔗 Hybrid Recommendation System

Combine collaborative filtering with content-based filtering for richer results.

🔍 Cold-Start Solution

Use genres, actors, and directors metadata to onboard new users with no history.

📡 Real-Time Streaming Pipeline

Live recommendation updates as users watch — event-driven architecture.

🌐 Context-Aware Recommendations

Factor in time of day, device type, and viewing session patterns.